

How Can You Think That?

Deliberation and the Learning of Opposing Arguments

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Abstract

Deliberation is commonly thought to increase people's awareness of both supporting and opposing arguments. But who learns, and what can an observational comparison tell us about that learning? We revisit a Deliberative Poll in Northern Ireland on the future of the school system. Its questionnaire asked respondents to list reasons that people supporting and opposing four policies might offer. One month after the event, participants gave an average of 7.90 coded reasons and contemporaneous controls gave 6.78. The difference is 1.12 reasons (95% confidence interval: -0.68 to 2.92). Adjustment for age, sex, religious community, and degree status yields a similar estimate, but the intervals are wide. Participation was self-selected, and the open-ended measure was not administered at the initial recruitment interview. The evidence therefore shows a positive but imprecise association, not an identified causal effect of deliberation. Participants did not differ detectably from controls in directional balance. Among the smaller complete-case panel, however, supporting and opposing repertoires became more balanced between the end of the event and the follow-up; coder-specific estimates are weaker. Even in this difficult setting, respondents could state reasons on both sides. Whether deliberation caused them to learn those reasons remains open.

Keywords: deliberation, political knowledge, opposing arguments,
Northern Ireland, Deliberative Polling

1 Introduction

In their everyday lives, many people have little idea how others holding policy preferences very different from theirs could do so [Mutz, 2002b]. The least attentive may not even recognize the existence of opposing opinions. Others may recognize their existence but never trouble to wonder how they could be held. Yet others may recognize their existence but be at a loss to explain them. And yet others may recognize them and attribute them to ignorance, stupidity, prejudice, or naked self-interest.

For that matter, some people—again, the least attentive—may have no conscious reason for their own policy preference. Some indeed have no real preference, and report “non-attitudes” in response to survey questions [Converse, 1964, 1970]. Others may have a preference and one or more conscious reasons for it. They may even recognize, without accepting, a few more. But many recognize only a few of the plausible reasons for their own views, and of course hold still fewer.

Deliberation may to some degree remedy all this. Deliberation, as the term is most commonly employed, is discussion, but not just any discussion. It is balanced, exposing its participants to competing arguments. It is informative, involving an exchange of information broadly construed. It is earnest, in the participants’ considering and weighing each other’s views while leaving open the possibility of changing their own. And it is civil.

The process should inject its participants’ everyday political reality with a good dose of additional cognition. It should increase their recognition and internalization of the possible reasons for their own existing preferences, enriching and possibly nuancing them, if not necessarily altering their broad thrust. That in itself can be normatively valuable insofar as it moves them closer, at least as a matter of degree or nuance, toward their considered or full-information views [de Tocqueville, 1969, Fishkin, 1991, Dahl, 1989, Arendt, 1968].

More distinctively, however, deliberation should increase participants’ awareness of the possible reasons for opposing preferences. That may be still more valuable, enriching and possibly nuancing their existing preferences—and sometimes altering their broad thrust when

a change would make them more considered. That is one possible benefit. The other is that participants who learn more of the other side’s reasons may regard them with greater respect, understanding better how others could legitimately, if mistakenly, disagree.

But who learns how many supporting and opposing arguments? How is that learning conditioned by what participants bring to the process? And what can the available comparison groups actually establish? We examine these questions using data from a Deliberative Poll on the future of the school system in the Omagh District Council area of Northern Ireland. The design does not warrant treating participant–control differences as effects of deliberation: respondents selected themselves into the event, and the study did not measure the open-ended outcome at recruitment. We therefore distinguish what the data describe from the causal account that motivated their collection.

2 Arguments

Let us begin by underscoring some conceptual distinctions. First, we are primarily concerned with the learning of reasons, not the persuasion or attitude change that may sometimes ensue. To cognize arguments is not necessarily to accept them, much less to accept enough of them to change one’s views. Participants in a deliberation may reject many, most, or all of the arguments they learn. What we examine is movement from “How could you think that?” to “I can see how you could think that” (and from “Why do I think that?” to “That’s why I think that”). In most cases, “I can see how you could think that” would in fuller form be followed by “although I still disagree.”

It is important to keep in mind that the “sides” here are sides of a policy debate, not sectarian groups. The question is the learning of arguments, especially those favoring the other side of the debate. This is not to say that identification with a sectarian group cannot affect learning. Where relations between groups are hostile, intergroup animus may lead some people to ignore arguments made by the opposing group, regardless of their content. If

the issue touches the sectarian cleavage, people may be especially unlikely to learn arguments about sharing resources or working together.

Prior work calls the relevant reasons people can give for their own and opposing opinions an argument repertoire [Cappella et al., 2002]. Some explicit terminology and notation make the argument clearer. For simplicity, assume a binary choice between policy options A and B , where B may simply mean not- A . An argument may be pro- or anti- A . Denote the numbers of pro- and anti- A arguments in long-term memory by n_p and n_a , their combined number by $n_c = n_p + n_a$, and their directional balance by

$$b_A = \frac{n_p - n_a}{n_p + n_a}. \quad (1)$$

If $n_a = 0$, then $b_A = 1$; if $n_p = 0$, then $b_A = -1$; and if $n_p = n_a$, then $b_A = 0$. Absolute imbalance is $|b_A|$. These quantities are implicitly indexed by person, topic, and measurement wave.

An argument's being "supporting" or "opposing" depends on the agreement or disagreement of its thrust with the individual's own position. Because we examine change over time, the relevant position is the initial one. Supporting arguments buttress that position; opposing arguments undermine it. Let s and o denote those counts. The total measured repertoire and its balance are

$$c = s + o, \quad b = \frac{s - o}{s + o}, \quad a = |b|. \quad (2)$$

Directional balance ranges from -1 , when every reason is opposing, to 1 , when every reason is supporting. Absolute imbalance a ranges from zero to one. Balance is undefined when $s + o = 0$; such respondent-waves do not enter analyses of b or a . For someone who initially favors a policy, pro-policy reasons are supporting and anti-policy reasons opposing. The labels reverse for someone who initially opposes it.

Parallel variables can characterize the arguments aired in the deliberation. Some may not reside in the head of any participant at the outset; most are presumably cognized by

some but by no means all. Denote the available pro- and anti- A arguments by N_p and N_a . These are outside-the-head entities residing in communications, not necessarily represented in every participant’s belief system.

2.1 Deliberation and the learning of arguments

At any moment, the number of pro- A arguments a person knows is a function of historical exposure to such arguments and historical incentives to absorb them. Illustratively, with no commitment to a particular functional form,

$$n_p = f(\alpha_0 + \alpha_1 N_p + \alpha_2 I_p), \quad n_a = f(\alpha_0 + \alpha_1 N_a + \alpha_2 I_a). \quad (3)$$

The parameters need not differ for pro- and anti- A arguments. The point is to distinguish exposure from attention and retention, a distinction long emphasized in work on information seeking and political communication [Atkin, 1973, Chaffee, 1972, Chaffee and McLeod, 1973].

H1: Deliberation increases the total number of cognized arguments. Deliberation should promote the learning of “new” arguments—perhaps previously encountered, but not previously cognized—both pro and con. It adds exposure and incentives: thinking about arguments is part of the task. Thus deliberation should increase both n_p and n_a , just as Deliberative Polls have often increased factual knowledge [Hansen, 2004, Luskin et al., 2009].

H2: In everyday life, cognized supporting arguments should outnumber cognized opposing arguments. People generally encounter and retain more attitudinally congenial than uncongenial considerations [Mutz, 2002b, Baron, 1998]. Adjacent research finds similar asymmetries for political facts and their interpretation [Delli Carpini and Keeter, 1991, 1997, Bartels, 2002, Shapiro and Bloch-Elkon, 2008, Jerit and Barabas, 2012, Gaines et al., 2007].

Part of this story is social exposure. People often live among others sharing their political orientations [Bishop and Cushing, 2009, Abrams and Fiorina, 2012, Cho et al., 2013, McDonald, 2011]. They also select associates with similar views [Verbrugge, 1977, 1983,

Huckfeldt and Sprague, 1995, Mutz and Martin, 2001, MacKuen, 1990, Bennett et al., 2000]. When they nevertheless interact with people holding opposing views, an interest in avoiding discord may keep disagreement off the table.

People also seek and select information that confirms existing views. The tendency applies to policy-relevant information and media choice [Iyengar et al., 2008, Garrett, 2009, Iyengar and Hahn, 2009, Stroud, 2008, Sunstein, 2001]. Once exposed, people process information less as neutral observers than as interested partisans [Chang, 2003, Lord et al., 1979]. Exposure, attention, and retention may therefore all favor supporting reasons.

H3: Deliberation narrows the everyday imbalance between supporting and opposing arguments. Deliberation should increase the total repertoire and may narrow its imbalance. The two claims are distinct. It airs arguments on both sides, more or less equally, but the newly encountered arguments need only be less imbalanced than the old repertoire. Suppose deliberation adds Δs supporting and Δo opposing reasons. Then

$$b' - b = \frac{2(o\Delta s - s\Delta o)}{(s + o)(s + o + \Delta s + \Delta o)}. \quad (4)$$

When the initial repertoire favors supporting reasons and all relevant quantities are positive, imbalance narrows if $\Delta s/\Delta o < s/o$: the new reasons need not be evenly divided, only less one-sided than the old repertoire. Total growth alone therefore says nothing about whether someone has become better able to articulate the opposing side.

Deliberation may supply just such a stream. It airs reasons on both sides and gives opposing reasons faces. A reason offered by a person one now knows is harder to ignore. Attention and retention may remain selective, but the new exposure and incentives should be less one-sided than those of everyday political life.

2.2 Conditioning factors

Other variables may condition these expectations. Anything that produces a scant initial store of arguments may also depress the number learned. Anything that accentuates the

initial imbalance creates more room for deliberation to level it. And anything that preserves unequal incentives to absorb supporting and opposing arguments may reduce that leveling.

H4: Policy attachments may dampen the learning of opposing arguments. Forced exposure does not eliminate selective attention and retention. People who favor A may be less likely to learn anti- A arguments, while those who oppose A may be less likely to learn pro- A arguments. Deliberation can therefore increase the total repertoire without producing symmetric gains or narrowing imbalance.

H5: Intergroup hostility may dampen both learning and the narrowing of imbalance. Suppose a population is divided into groups such as Catholics and Protestants. Dislike of the other group may limit attention to arguments associated with its members. The problem is most acute when group membership is correlated with position on the issue: members of each group may then tune out the arguments disproportionately offered by the other. On issues expressly involving group rights, resources, or contact, intergroup affect may also spill directly into the policy judgment [Altemeyer, 1996, Marcus et al., 1995, Kim, 2011]. Yet these conditions need not all hold. If Catholics and Protestants take similar positions on an issue, participants can hear both sides from members of both groups.

H6: Deliberation may produce a larger increase among those who already know more. Participants arriving with more arguments may learn still more, just as those arriving with more factual knowledge often acquire more in information-rich settings. Existing knowledge can provide hooks for organizing new material rather than crowding it out. This expectation sits against the generally low-information environment of mass politics, in which acquiring and organizing policy information is costly and uneven [Downs, 1957, Iyengar, 1990, Gilens, 2001].

H7: Learning, especially of opposing arguments, may be greater in more attitudinally diverse groups. Sociodemographic diversity need not matter except insofar as it predicts policy

attitudes. But encountering people with opposing positions should increase exposure to their reasons. At the group level, that opportunity is greatest when the group is divided rather than unanimous.

These seven propositions reproduce the theoretical setup that motivated the original study. The available design, however, does not identify all seven as causal hypotheses. The revised analysis directly estimates repertoire size and directional balance. Policy attachment, intergroup hostility, prior knowledge, and group composition remain mechanisms for a design with baseline repertoire measurement and verified random group assignment. The algebra organizes the estimands; it is not offered as a new formal model.

3 The Omagh Deliberative Poll

The Deliberative Poll took place in Omagh, a district council area in County Tyrone, Northern Ireland, with nearly 48,000 inhabitants at the time. Omagh had a mixed population of Catholics and Protestants and primary and post-primary schools representing all the major school types. Some schools had mainly Catholic pupils, some mainly Protestant pupils, and a few a more even mix. As in Northern Ireland as a whole, the birthrate and school-age population were declining, forcing schools to consider greater coordination or consolidation [Department of Education, Northern Ireland, 2007].

The setting was a classic deeply divided society with a still-recent history of violent strife. The Protestant–Catholic divide was reinforced by residential segregation, divided public housing, segregated schooling, endogamy, and historically distinct employment patterns [Boyle and Hadden, 1994, O’Hara, 2004, Cohen, 2007, Pogatchnik, 2008]. From the standpoint of those skeptical of mass deliberation across deep divisions, bringing randomly sampled parents together to discuss schools was itself consequential. At the same time, Catholics and Protestants did not divide neatly on every policy considered here, so sectional hostility need not translate directly into refusal to learn a policy argument. A companion

account examines this event and deliberation across the community divide in detail [Luskin et al., 2014].

The event followed the standard Deliberative Poll design [Fishkin, 1991, Luskin and Fishkin, 2002, Luskin et al., 2002, Fishkin, 2009]. A random sample of parents of school-aged children was interviewed at recruitment and invited to a long day of deliberation. Of the 568 initial interviewees, 124 took part. Participants received balanced briefing materials and met in moderated small groups, alternating with plenary sessions in which they questioned policy experts and representatives of the different school types. Participants completed a survey at the end of the event. A separate control group was interviewed later, together with a subset of participants roughly one month after the event. The corresponding analytic counts before coding exclusions are 150 controls and 93 returning participants.

Like other Deliberative Polls, the Omagh event brought a diverse set of people together in an environment intended to support a respectful airing of both similarities and differences. In expectation, participants encountered common briefing materials and the same plenary program, while their small-group discussions supplied varying mixes of facts, experiences, and reasons. Those ingredients motivate the learning hypotheses, but they do not make every participant's experience uniform.

Nor do Northern Ireland's deep divisions necessarily pose the same obstacle for every policy. Let A be a proposal involving religious mixing. Hostility toward the other community may be one reason to oppose A , and an opponent may initially possess many more anti- A than pro- A reasons. During deliberation, that person could ignore pro- A arguments associated with the other community. But the process could also make those arguments more difficult to ignore, especially when similar arguments come from members of the person's own community. The direction of the resulting change is therefore an empirical question, not something the social cleavage settles by itself.

This design contains both an experiment and an observational comparison, and they should not be conflated. Participants were reportedly assigned to small discussion groups at random. If that assignment is verified, contrasts among group compositions may support

causal claims about those compositions. Attendance at the event was not randomly assigned: invitees chose whether to attend. The participant-control contrast can therefore reflect pre-existing differences, event participation, briefing exposure, measurement timing, or any combination of them. The original setup supplies the mechanisms and expectations; the revised empirical sections state more narrowly what these comparisons can establish.

4 Measurement

4.1 Open-ended questions

The post-event and follow-up surveys contained paired open-ended questions about four school policies: all-ability schools, balanced religious enrolment, classroom collaboration between schools of different religious composition, and combining primary with post-primary pupils. For each policy, respondents could list up to five reasons that strong supporters might offer and up to five reasons that strong opponents might offer. Closed-ended policy questions allow us to classify each elicited reason as supporting or opposing the respondent's initial position.

For all-ability schools, respondents first rated the option of “having a system of all-ability schools, all providing the same wide curriculum.” They were then asked for up to five reasons that people who strongly supported the proposal would give and up to five reasons that people who strongly opposed it would give.

For combining schools, respondents rated the option of schools combining primary and post-primary pupils—for example, ages 7–14. The follow-ups again asked separately for the reasons strong supporters and strong opponents would give.

For collaboration across schools, respondents rated whether, when schools of different religious composition entered partnerships, children from both schools should at least sometimes be taught in the same classroom. The paired open-ended questions asked for the reasons of those who strongly agreed and those who strongly disagreed.

For balanced enrolment, the closed-ended item ran from the view that children should attend school only with children of their own religion to the view that they should attend schools with a balanced enrolment of Protestant and Catholic pupils. Respondents then listed reasons for each endpoint. These prompts asked respondents to reproduce arguments that people on each side might offer; they did not ask respondents to endorse those arguments.

Two coders, blind to the hypotheses, classified each response by its substance. A third coder was assigned to adjudicate disagreements. The source file stores a label set from each coder, because one response could contain more than one reason. The coding pipeline parses those sets explicitly. It regards reordered sets as agreements, uses the adjudicator when the first two coders disagree, and leaves a disagreement unresolved when no adjudication is present. Normalized exact agreement is 92.0%, and nominal Krippendorff’s α is 0.79. There are 89 unresolved response slots. These reliability statistics describe the reproducibility of the coding decisions; they do not validate the construct itself [Krippendorff, 1980, 1986, 2004].

The primary outcome counts response slots that contain at least one substantive reason. Codes for nonsensical, missing, or vague answers are excluded. On prompts where saying that the other side is merely prejudiced gives that side too little credit, that response is also excluded. An inclusive version retains those responses. Further sensitivities count every coded mention and use each original coder separately. If any of a respondent’s slots remains unresolved, the primary total is missing; the coder-specific sensitivities retain the respondent.

Directional outcomes sum supporting and opposing reasons across policies for which the respondent has a nonneutral initial position. We calculate b and $|b|$ only when both sums are observed and their combined total is positive. This complete-case rule keeps an unresolved slot from being silently treated as no reason. Because that rule reduces the paired sample, we also report the balance estimates separately for each original coder.

4.2 Other measures

The questionnaires also measured factual knowledge of the school system, policy attitudes, religious affiliation, education, age, and sex. They included feeling ratings toward Catholics and Protestants and trait ratings of whether members of each community were trustworthy and open to reason. The original study used these variables to motivate analyses of selection, intercommunity affect, prior knowledge, and group composition.

For the revised primary comparison, age is measured in years; sex is represented by an indicator for women; religious community distinguishes Catholic and Protestant respondents; and degree status records whether the respondent reports a bachelor’s degree or higher. The Supporting Information shows these variables across recruitment, attendance, return, and control samples. The primary adjustment uses them to describe measured composition, not to claim that selection has been eliminated.

The original group-level argument used the mean factual knowledge of the other participants in the discussion group and an index of attitudinal heterogeneity across the four policy items. The latter follows the variance–covariance construction described by Anderson [2003]. Those quantities remain conceptually relevant to H6 and H7. We do not treat their associations with post-event responses as causal here because the documentation available in the repository does not independently verify the small-group randomization or support the original multilevel specification. They are therefore mechanisms for future analysis rather than parts of the primary claim.

4.3 Analytic comparisons

The primary estimand is the respondent-level mean difference in the total number of coded reasons between returning participants and contemporaneously interviewed controls. We report the raw difference and an ordinary least squares adjustment for age, sex, religious community, and degree status. The adjustment addresses measured composition only. It does not turn the comparison into an experiment.

Two comparisons are secondary. First, we compare each returning participant’s end-of-event and follow-up responses. The paired change describes retention or decay among returnees, but it also combines changes in effort, survey context, and mode. Second, we compare all end-of-event participants with later controls. That contrast has a larger sample but is not contemporaneous and includes any learning from the briefing materials before the end-of-event survey.

These comparisons face several distinct threats. Attendance was voluntary: of the people who completed the recruitment interview and were invited, most did not attend. Some participants then failed to return the later questionnaire. Sampling variation, selection into attendance, and attrition can each alter the composition of the samples. The available covariates reveal some of that composition but cannot establish comparability on motivation, political interest, verbal fluency, or willingness to complete open-ended questions. Spillover is also possible if members of the later control sample encountered discussion or materials connected with the event. Finally, a Deliberative Poll is a collection of parallel small-group experiences rather than one perfectly uniform treatment.

The original paper described the T2-control and T3-control contrasts as treatment effects. That wording was too strong. The event invitation did not randomize attendance, and the open-ended outcome was not measured at recruitment. The revised analysis calls these sample differences and reserves causal language for a design that identifies the relevant contrast. For the same reason, we do not repeat the original mediation analysis. A formal mediation estimator [Imai et al., 2010] cannot rescue a treatment effect that the research design does not identify.

For the directional outcomes, we estimate the same contemporaneous T3 contrast and paired T2-T3 change. The supporting and opposing totals show which side changes. Directional balance shows the net tilt toward a respondent’s initial position, while absolute imbalance asks only how far the repertoire lies from parity.

The unit of analysis is the respondent. Participants may share shocks induced by their moderated discussion group, while controls were not grouped. Confidence intervals there-

fore use CR2 standard errors, clustering participants by their observed discussion group and treating controls as singleton clusters, with Satterthwaite degrees of freedom. This adjustment addresses dependence; it does not repair the observational comparison. The analysis is unweighted because the available files do not contain a documented sampling weight for these comparisons. All analytical values in the manuscript are generated by the R pipeline.

5 Results

5.1 Total reasons

Figure 1 shows the respondent-level means. Returning participants at follow-up gave 7.90 coded reasons on average, compared with 6.78 among controls. The unadjusted difference is 1.12 reasons (95% confidence interval: -0.68 to 2.92; $p = .215$). The adjusted difference is 0.97 reasons (95% confidence interval: -0.83 to 2.77; complete-case $N = 174$). Both estimates are positive, but both intervals include zero and differences of substantively meaningful size in either direction. On a multiplicative scale, participants gave 16.5% more reasons than controls (count ratio 1.17, 95% confidence interval: 0.92 to 1.48). The ratio is useful for magnitude; it remains an associational comparison.

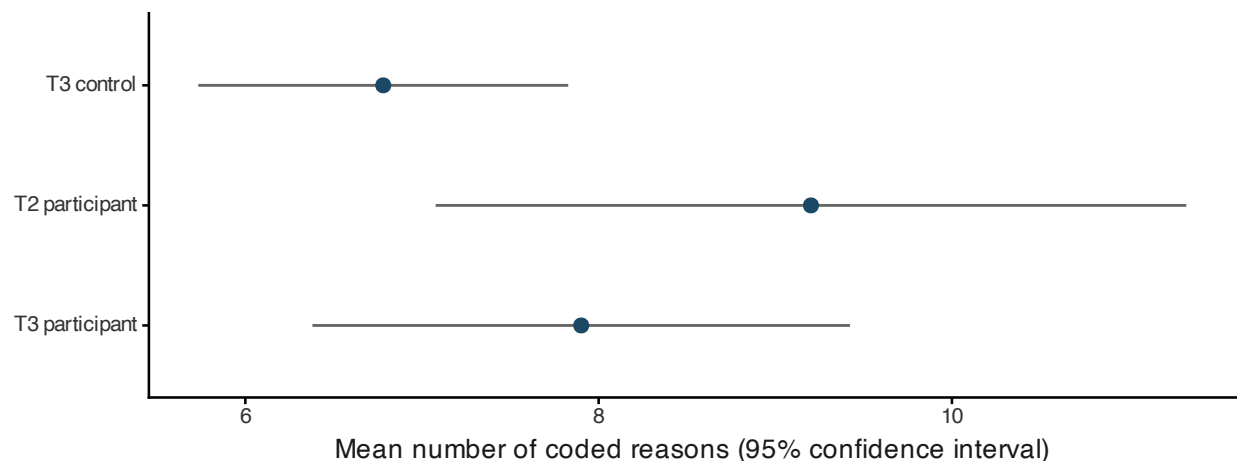


Figure 1: Mean number of coded reasons by sample and wave. Points are respondent-level means; lines are 95% CR2 confidence intervals using participant discussion groups and singleton control clusters. The primary coding excludes invalid and vague responses, question-specific answers that give the other side too little credit, and respondent waves containing an unresolved coder disagreement. The T2 participant and T3 control samples were not interviewed contemporaneously.

The paired estimate is also imprecise. Among returnees with resolved coding at both waves, the follow-up total is -0.72 reasons lower than the end-of-event total (95% confidence interval: -3.26 to 1.82). The noncontemporaneous end-of-event participant difference is larger: 2.42 reasons (95% confidence interval: 0.15 to 4.70). That pattern is consistent with short-run exposure or effort followed by decay, but the design cannot distinguish that account from selection and measurement differences.

Table 1: Primary and secondary respondent-level comparisons

Panel	Sample or contrast	Quantity	Estimate [95% CI]	<i>N</i>
Means	T3 control	Mean	6.78 [5.73, 7.83]	114
Means	T2 participant	Mean	9.20 [7.08, 11.33]	104
Means	T3 participant	Mean	7.90 [6.38, 9.42]	71
Comparisons	T3 participant minus T3 control, unadjusted	Difference	1.12 [-0.68, 2.92]	185
Comparisons	T3 participant minus T3 control, adjusted	Difference	0.97 [-0.83, 2.77]	174
Comparisons	Participant change, T3 minus T2	Difference	-0.72 [-3.26, 1.82]	61
Comparisons	T2 participant minus T3 control	Difference	2.42 [0.15, 4.70]	218
Comparisons	T3 participant divided by T3 control	Mean ratio	1.17 [0.92, 1.48]	185

Note: Differences are in coded reasons. Intervals use CR2 standard errors with Satterthwaite degrees of freedom. Participant observations are clustered by discussion group; controls are singleton clusters. The adjusted T3 comparison includes age, sex, religious community, and degree status. The paired row compares the same returning participants. The last row compares samples interviewed at different times.

5.2 Supporting, opposing, and balanced repertoires

Table 2 separates repertoire size from direction. At T3, participants gave slightly more supporting and opposing reasons than controls. But their directional balance differs by only -0.01 (95% confidence interval: -0.14 to 0.12; $p = .871$), and their absolute imbalance by -0.04 (95% confidence interval: -0.15 to 0.06). The participant–control association in total repertoire is therefore not concentrated in a clear shift toward the opposing side.

The paired comparison is different. Among returnees with complete primary coding and a positive repertoire at both waves, directional balance changes by -0.22 (95% confidence interval: -0.42 to -0.02; $p = .030$), while absolute imbalance changes by -0.20 (95% confidence interval: -0.33 to -0.08). In this complete-case panel, repertoires move toward parity. Most of the movement comes from fewer supporting reasons rather than more opposing reasons,

so it is evidence of narrowing, not evidence that respondents accumulated opposing reasons during the month.

Table 2: Supporting and opposing argument repertoires

Outcome	T3 control	T3 participant	T3 difference [95% CI]	<i>N</i> (T3)	Paired change [95% CI]	<i>N</i> (paired)
Supporting reasons	3.40	3.71	0.32 [-0.57, 1.21]	211	-1.10 [-2.15, -0.05]	72
Opposing reasons	2.38	2.51	0.13 [-0.46, 0.72]	226	-0.35 [-1.18, 0.47]	82
Directional balance (<i>b</i>)	0.23	0.22	-0.01 [-0.14, 0.12]	176	-0.22 [-0.42, -0.02]	51
Absolute imbalance (<i> b </i>)	0.37	0.33	-0.04 [-0.15, 0.06]	176	-0.20 [-0.33, -0.08]	51

Note: Supporting and opposing are defined relative to each respondent’s initial policy position. Counts sum reasons across classifiable policies. Directional balance is $b = (s - o)/(s + o)$, and absolute imbalance is $|b|$; both are undefined when $s + o = 0$. Differences use CR2 standard errors with Satterthwaite degrees of freedom. T3 differences are participant minus control; paired changes are T3 minus T2 among returning participants. Sample sizes vary because unresolved coding, neutral positions, and zero repertoires affect the four outcomes differently.

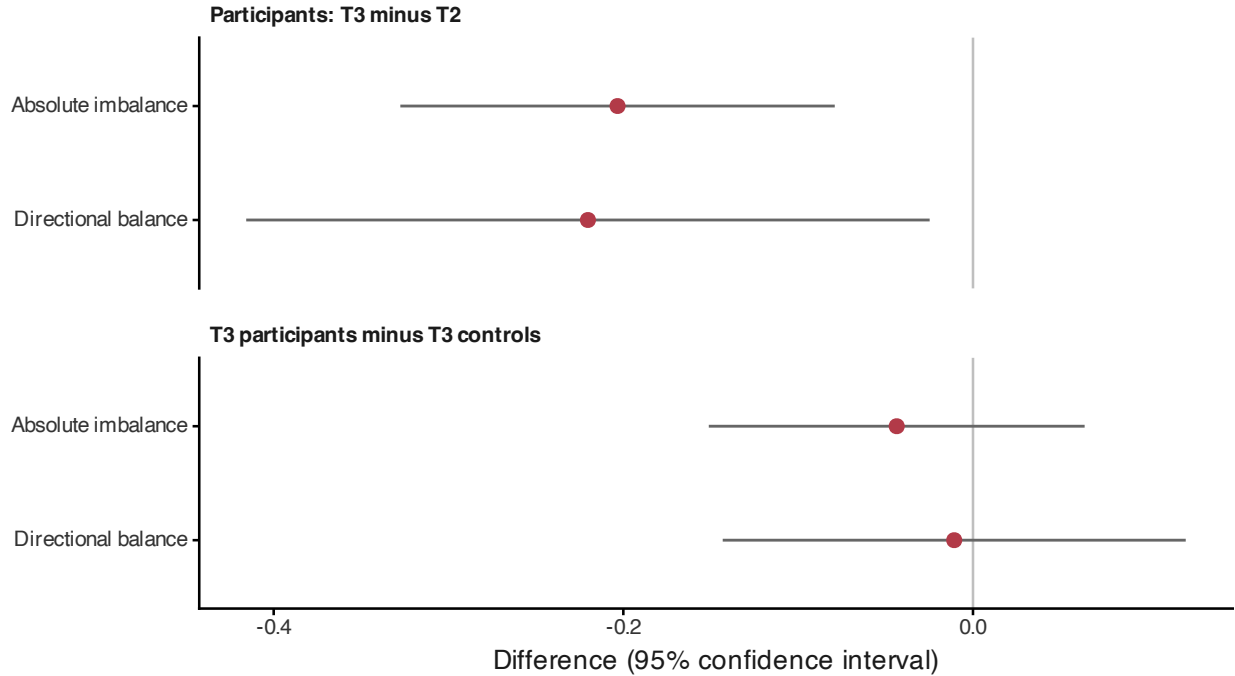


Figure 2: Changes in directional balance and absolute imbalance. Points show the T3 participant–control difference and the paired T3–T2 change; lines are 95% CR2 confidence intervals. Zero denotes no difference or change. Supporting reasons raise b and opposing reasons lower it; smaller $|b|$ means a more even repertoire.

5.3 Sensitivity to coding choices

The inclusive count, the count of coded mentions, and the two coder-specific totals all yield positive T3 participant–control differences. Their intervals overlap the primary interval and generally include zero. The coder-specific estimates use the full contemporaneous sample and show that dropping respondent waves with unresolved adjudication reduces precision without

creating the direction of the result. These sensitivity intervals use the same partially clustered CR2 procedure as the primary estimate. The paired balance result is more sensitive: both coder-specific changes point toward greater balance, but they are smaller than the primary complete-case estimate and their 95% intervals include zero. The Supporting Information reports every specification.

6 What the evidence establishes

The strongest result is descriptive. Returning participants could articulate reasons on both sides of contentious school-policy debates in a deeply divided setting. They gave somewhat more valid reasons than contemporaneous controls, but the estimate is imprecise after the coding rules are enforced. The data do not establish that deliberation caused the difference.

Participation was self-selected, and the open-ended outcome was not measured in the initial recruitment interview. Covariate adjustment can compare respondents who look similar on recorded demographics, but it cannot remove unmeasured interest, motivation, verbal fluency, or willingness to spend a day at the event. The later control interview also differs from the participant surveys in timing and experience. These are properties of the design, not statistical inconveniences that a more elaborate model can solve.

The paired comparison answers a narrower question. It follows the same people from the end of the event to the later survey, so stable differences among people do not confound the change. But the end-of-event survey came after briefing materials and the day's proceedings, and the follow-up took place under a different context. The decline is compatible with forgetting; it is also compatible with lower effort or a changed response situation. The same qualification applies to the apparent narrowing of directional imbalance. Because supporting reasons decline more than opposing reasons, the result need not reflect new understanding of the opposing side.

The measurement analysis adds a caution. A response can contain multiple reasons, coders can disagree about its labels, and the primary count requires a decision about whether

each response contains a substantive reason. Explicit set parsing, adjudication states, and coder-specific sensitivities make those decisions visible. The similarity of the sensitivity estimates is reassuring for total repertoire size, but the directional paired estimate moves materially when each original coder is used. Coder agreement cannot establish that the count captures every relevant kind of argument equally well.

7 Discussion

The motivating idea remains important. Deliberation may enable movement from “How could you think that?” to “I can see how you could think that,” even when disagreement remains. The Omagh respondents demonstrate that people in a deeply divided society can state arguments on both sides of policies touching sectional lines. But a capacity demonstrated in a selected participant sample is not the same as an average causal effect of deliberation.

Earlier research found that people who encounter cross-cutting political networks tend to be more aware of arguments on the other side, though such networks can also discourage political participation [Mutz, 2002b,a]. Yet those who choose such exposure may differ in unmeasured ways from those who do not. Everyday discussion may sometimes overcome imbalance when it is sustained, engaging, and sufficiently broad. A structured deliberative event is designed to make those conditions less exceptional. The remaining empirical problem is to separate what the event caused from the characteristics of the people who attended it.

This study also remains one Deliberative Poll on one topic. Northern Ireland’s divisions affect many features of its mass and elite politics, including what participants bring to a discussion about schools. That makes Omagh a demanding and substantively important setting, but it does not by itself reveal whether the same pattern would appear for other issues, populations, or forms of deliberation.

One conception of considered preferences asks what people would think if they were well informed, heard competing opinions, and understood the consequences of the alternatives [Fishkin, 1991, 2009, Dahl, 1989, Arendt, 1968]. Most people know and have considered

relatively little about most policy questions. Deliberative Polling seeks to provide factual information and an opportunity to encounter competing reasons. Learning an opposing argument may therefore matter even when it does not change a policy preference: continued disagreement can become more informed and more intelligible.

A stronger study would measure the open-ended outcome before invitations or briefing materials, randomize access to the deliberative event among willing respondents, and repeat the same measurement after the event and at follow-up. It would preserve the actual arguments, their coder label sets, and all adjudications in a tidy long format. Such a design would separate selection, immediate learning, and retention. It would also permit a direct test of whether gains are concentrated among opposing rather than supporting reasons.

For now, the evidence supports a narrow conclusion. Returning participants articulated somewhat more reasons than controls, and both groups could articulate opposing reasons. The T3 comparison shows no detectable difference in directional balance. The paired complete-case estimate moves toward balance, but its sensitivity to coder choice and its source in declining supporting reasons limit what it establishes. The uncertainty is too large, and the design too confounded, to attribute these patterns to deliberation. This is not a null conclusion about deliberation. It is the question the available evidence can answer.

Detailed sample diagnostics, policy-specific estimates, model expansions, coding sensitivities, and subgroup descriptions appear in the Supporting Information. The build starts from the pinned source files in `data/manifest.yaml`; the R pipeline generates every analytical value, table, and figure in this paper.

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Supporting Information

The following tables report expanded descriptive and sensitivity results. All tables are generated by the analysis pipeline. Estimates use the primary coding unless a table states otherwise. Confidence intervals use CR2 standard errors with Satterthwaite degrees of freedom, clustering participants by discussion group and treating controls as singleton clusters.

S.1 Sample composition

Table S1: Composition of recruitment, participant, returnee, and control samples

Sample	<i>N</i>	Age	Women (%)	Catholic (%)	Degree (%)
T1 nonparticipant	444	40.0 (436)	69.1 (444)	72.6 (424)	18.8 (441)
T2 participant	124	40.7 (124)	75.8 (124)	65.8 (120)	19.5 (123)
T3 nonreturning participant	31	37.9 (31)	74.2 (31)	66.7 (30)	12.9 (31)
T3 returning participant	93	41.6 (93)	76.3 (93)	65.6 (90)	21.7 (92)
T3 control	150	40.0 (149)	72.7 (150)	73.9 (142)	18.7 (150)

Note: Cells contain a mean or percentage followed by the nonmissing denominator in parentheses. T1 nonparticipants completed the recruitment interview but did not attend the event. T3 nonreturning participants attended but did not complete the follow-up. T3 controls were recruited separately.

S.2 Policy-specific results

Table S2: Mean coded reasons and contrasts by policy and prompt

Policy	Prompt	T2 P	T3 P	T3 C	T2 P – T3 C	T3 P – T3 C	T3 – T2
All-ability schools	Pro-policy	1.71 (118)	1.19 (89)	1.00 (147)	0.71 [0.31, 1.12] (265)	0.19 [-0.11, 0.50] (236)	-0.56 [-1.03, -0.10] (85)
All-ability schools	Anti-policy	1.19 (119)	0.85 (92)	0.83 (145)	0.37 [-0.03, 0.77] (264)	0.02 [-0.24, 0.28] (237)	-0.43 [-0.91, 0.06] (89)
Balanced enrolment	Anti-policy	1.49 (123)	1.13 (92)	0.98 (146)	0.51 [0.22, 0.79] (269)	0.15 [-0.13, 0.43] (238)	-0.30 [-0.64, 0.04] (92)
Balanced enrolment	Pro-policy	1.88 (118)	1.60 (78)	1.26 (127)	0.62 [0.19, 1.05] (245)	0.34 [0.01, 0.67] (205)	-0.20 [-0.61, 0.21] (74)
Inclusive classrooms	Pro-policy	1.37 (124)	1.10 (92)	1.01 (147)	0.36 [0.01, 0.72] (271)	0.09 [-0.19, 0.37] (239)	-0.30 [-0.69, 0.08] (92)
Inclusive classrooms	Anti-policy	0.64 (121)	0.49 (93)	0.57 (150)	0.07 [-0.18, 0.32] (271)	-0.08 [-0.29, 0.13] (243)	-0.12 [-0.49, 0.25] (90)
Combined primary and post-primary schools	Pro-policy	0.92 (123)	1.02 (93)	0.72 (149)	0.19 [-0.08, 0.47] (272)	0.30 [0.04, 0.55] (242)	0.11 [-0.25, 0.47] (92)
Combined primary and post-primary schools	Anti-policy	0.79 (122)	0.92 (91)	0.81 (149)	-0.02 [-0.35, 0.31] (271)	0.12 [-0.15, 0.38] (240)	0.16 [-0.25, 0.56] (90)

Note: T2 P and T3 P denote participants at the end of the event and follow-up; T3 C denotes contemporaneous controls. The first two differences compare distinct samples. T3 minus T2 is a within-returnee change. Contrast cells contain an estimate and 95% confidence interval. Prompt labels refer to the policy direction, not to the respondent's own position. Parentheses contain the sample size for each mean or contrast.

S.3 Response-slot order

Table S3: Linear probability model for whether a response slot contains a valid reason

Term	Estimate	CR2 SE	CI low	CI high	Rows	Clusters
T3 participant	0.076	0.036	0.002	0.149	9656	170
Response-slot order	-0.130	0.006	-0.143	-0.118	9656	170
Participant $\times$ response-slot order	-0.024	0.010	-0.044	-0.005	9656	170

Note: The T3 slot-level model includes policy and prompt-side fixed effects. Response slot order runs from zero for the first offered slot to four for the fifth. The participant coefficient is therefore the participant–control difference at the first slot; the interaction shows how that difference changes across later slots. The model describes response production, not a causal effect of participation.

S.4 Coding sensitivity

Table S4: Sensitivity of the total-repertoire T3 comparison to coding choices

Outcome construction	T3 difference [95% CI]	N	Clusters
Inclusive low-credit coding	1.00 [-0.83, 2.84]	185	134
Count every coded mention	1.15 [-0.65, 2.96]	185	134
Coder 1	1.28 [-0.23, 2.78]	243	170
Coder 2	1.08 [-0.38, 2.55]	243	170

Note: Estimates are participant minus control. Inclusive coding retains question-specific low-credit answers. The mention count credits each distinct code rather than each occupied response slot. Coder-specific outcomes do not require adjudication and therefore retain more respondents.

Table S5: Sensitivity of directional balance to coding choices

Construction	T3 difference [95% CI]	N (T3)	Paired change [95% CI]	N (paired)
Inclusive coding: b	0.03 [-0.10, 0.15]	177	-0.20 [-0.34, -0.06]	51
Inclusive coding: $ b $	-0.04 [-0.13, 0.05]	177	-0.17 [-0.27, -0.07]	51
Coder 1: b	-0.01 [-0.12, 0.10]	221	-0.14 [-0.29, 0.01]	78
Coder 1: $ b $	-0.04 [-0.14, 0.05]	221	-0.10 [-0.21, 0.02]	78
Coder 2: b	0.03 [-0.09, 0.14]	221	-0.12 [-0.25, 0.01]	78
Coder 2: $ b $	-0.02 [-0.11, 0.08]	221	-0.07 [-0.16, 0.02]	78

Note: Directional balance is $b = (s - o)/(s + o)$ and absolute imbalance is $|b|$. Balance is undefined for a zero repertoire. T3 differences are participant minus control; paired changes are T3 minus T2. Inclusive coding retains low-credit responses but still requires completed adjudication. Coder-specific outcomes retain unadjudicated cases.

S.5 Religious-community descriptions

Table S6: T3 total repertoires by religious community

Community	Control mean	<i>N</i> control	Participant mean	<i>N</i> participant	Difference [95% CI]	<i>N</i> difference
Catholic	7.33	85	7.77	43	0.44 [-1.97, 2.84]	128
Protestant	4.82	22	7.88	26	3.07 [0.81, 5.32]	48

Note: Differences are participant minus control. Respondents outside the two largest religious communities and respondents with missing affiliation are omitted. These subgroup estimates are descriptive, were not selected as separate confirmatory tests, and should not be read as evidence of effect heterogeneity.

S.6 Coding rules and reproducibility

The substantive categories follow the original codebook. Codes for “other” remain valid; codes for nonsensical, missing, and vague responses do not. For three prompts, code three attributes the opposing position to prejudice or ignorance rather than stating a policy reason. The primary measure excludes that code; the inclusive sensitivity retains it.

Coder labels are normalized to sorted integer sets before agreement is evaluated. If both coders provide the same set, that set is final. If they differ and the adjudicator provides a set, the adjudicated set is final. If they differ without adjudication, the primary respondent- wave outcome is unresolved. Coder-specific sensitivities preserve those observations.

The numbered R scripts create the analysis data, one bundled results object, headline CSV files, tables, figures, and manuscript macros. Running `make check` lints the code, rebuilds every output, runs the validation tests, and compiles this PDF.